

Initial Experiments: Extending the MVP Simulator

William Dennis

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Extends the MVP simulator (`docs/mvp/mvp_specification.tex`) by enriching the state space, action space, and reward function, moving toward the full MDP defined in `docs/design/initial_design.tex`.

1 State Space Extension

References `docs/design/initial_design.tex` §3 and Issue #124.

1.1 Enriched StateView

1.2 Contextual Bandit Conditioning

1.3 Open Questions

2 Action Space Extension

References `docs/design/initial_design.tex` §3 and Issue #126. *Implementation: PR 1a (feat/1a-actions) — merged.*

2.1 ActionConfig

The `actions` field in `MDPConfig` accepts two formats:

- `list[str]`: MVP mode (backward compatible), e.g. `[nudge, idle]`.
- `list[dict]`: Extended mode with per-action metadata (`name`, `reward_penalty`, `burden_penalty`).

The `_normalize_actions()` validator extracts names into a flat list, rejects mixed types, duplicates, and missing `name` keys. Agents receive `config.action_names` (always `list[str]`), so no ABC changes are needed.

2.2 Transition Model Updates

The 6-action config requires a 6×2 transition matrix (6 actions, 2 states). The transition probabilities are hand-tuned placeholders in `docs/initial_experiments/configs/1a_actions_only.yaml`.

2.3 Agent Compatibility

All 9 MVP agent variants work with the 6-action config. The `action_names` property returns a flat `list[str]` regardless of input format, so agents require no changes.

2.4 Open Questions

- Penalty values are placeholders. A grid search could identify meaningful regions where burden penalty shapes behaviour.
- With 6 actions and 2 states, the transition matrix has 12 entries. Real clinical data would inform realistic probabilities.

3 Reward Function Extension

References docs/design/initial_design.tex §3 and Issue #132.

3.1 Multi-Timescale Reward

3.2 Reward Composition

3.3 Synthetic Clinical Data

3.4 Open Questions

4 Combined Results